# Particle Swarm Optimization

How individual memory and social information balance exploration and exploitation

CS420 Introduction to AI · Seminar S01 · November 2026

[Member 1] · [Member 2] · [Member 3]

# Why particle swarms?

Member 1

- Minimise a function with no usable gradient
- Local search stops at the first basin
- PSO: many particles that share what they find

**Guiding question**

How does PSO combine individual experience and social information to balance exploration and exploitation?

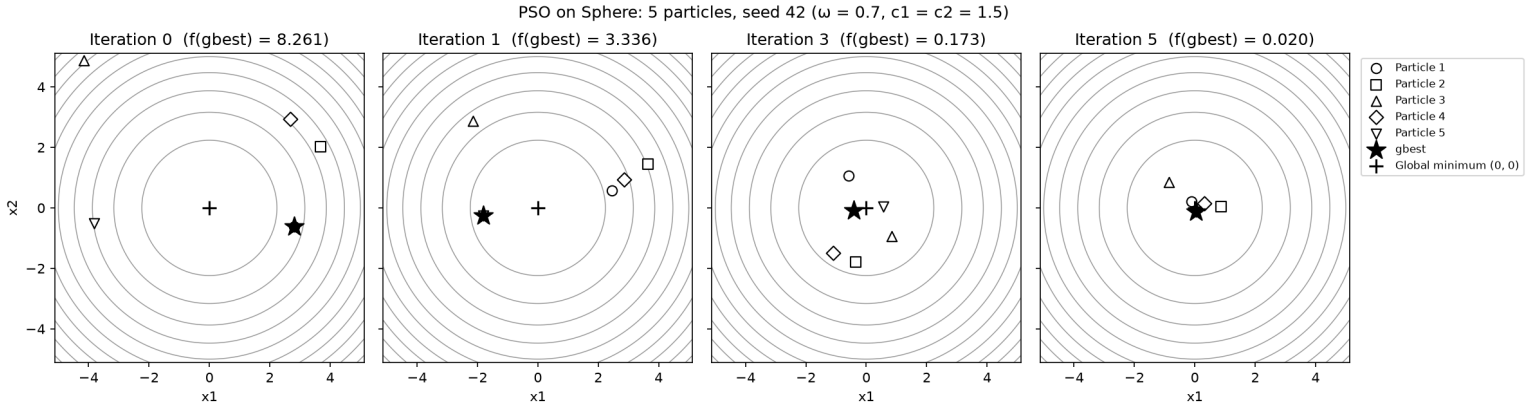



*Figure: five particles converging on Sphere (team experiment)*

# From flocking to an optimiser

Member 1

$$v \leftarrow v + 2 \cdot \text{rand}() \cdot (\text{pbest} - x) + 2 \cdot \text{rand}() \cdot (\text{gbest} - x)$$

1. **pbest**
   Each agent remembers its own best position ("simple nostalgia")
2. **gbest**
   Best position known to everyone (a group norm)
3. **× 2**
   Mean pull of 1, so agents overshoot about half the time

**Unbalanced pulls fail:** a much stronger pull to gbest made the flock rush prematurely towards local minima; a stronger pull to pbest made agents wander. Removing the old velocity was "quite ineffective at finding global optima".

*Kennedy & Eberhart (1995), pp. 1945–1946*

# What changed by 2007

Member 1

| | Kennedy & Eberhart (1995) | Poli et al. (2007) |
|---|---|---|
| **Velocity control** | Fixed coefficient 2, no damping | Inertia ω (1 − ω acts as friction) or constriction χ |
| **Recommended setting** | Not studied | $\chi \approx 0.7298 \Leftrightarrow \omega = 0.7298$, $c_1 = c_2 \approx 1.496$ |
| **Who informs a particle** | Whole swarm (gbest) | gbest is fast; lbest ring resists local optima |
| **Evidence** | A few demonstrations | About 700 application papers reviewed |

**Our default: $\omega = 0.7$, $c_1 = c_2 = 1.5$**

A rounded version of the standard constriction setting. High ω explores, low ω exploits; we test 0.4, 0.7 and 0.9.

*Poli, Kennedy & Blackwell (2007), Sects. 2.2–2.4, 3.1, 6*

# Problem, functions and update rule

Member 2

**Sphere**

$f(x) = \Sigma\, x_i^2$

One basin: tests convergence speed

**Rastrigin**

$f(x) = 20 + \Sigma\, (x_i^2 - 10 \cos 2\pi x_i)$

Grid of local minima: tests trapping

*Both in $[-5.12, 5.12]^2$, global minimum 0 at (0, 0); fitness = f(x), lower is better*

$$v \leftarrow \omega\, v + c_1 r_1 (pbest - x) + c_2 r_2 (gbest - x), \quad x \leftarrow x + v$$

inertia · cognitive: own best · social: swarm best

**2,020**

evaluations per run: 20 particles × 100 iterations + 20 at start; $|v| \le 2$; $r_1$, $r_2$ new for every particle, dimension and iteration

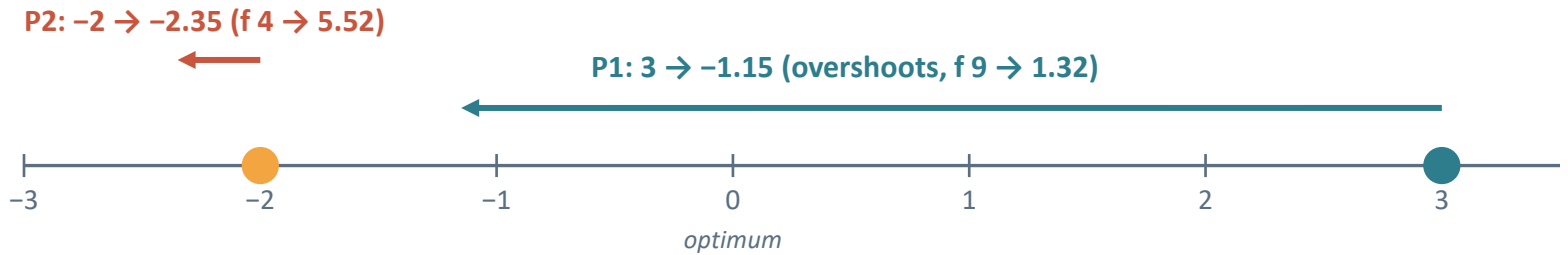
*Update rule: Poli et al. (2007), Eq. 2. Functions and settings: team design*

# One step by hand

Member 2

$f(x) = x^2$, $\omega = 0.7$, $c_1 = c_2 = 1.5$; start: $x_1 = 3$, $x_2 = -2$ (gbest)

| Particle | Inertia | Cognitive | Social | v(1) | x(1) | f(x(1)) |
|---|---|---|---|---|---|---|
| 1 | 0.35 | 0 | 1.5·0.6·(−5) = −4.5 | **−4.15** | **−1.15** | 1.3225 |
| 2 (gbest) | −0.35 | 0 | 0 | −0.35 | −2.35 | 5.5225 |

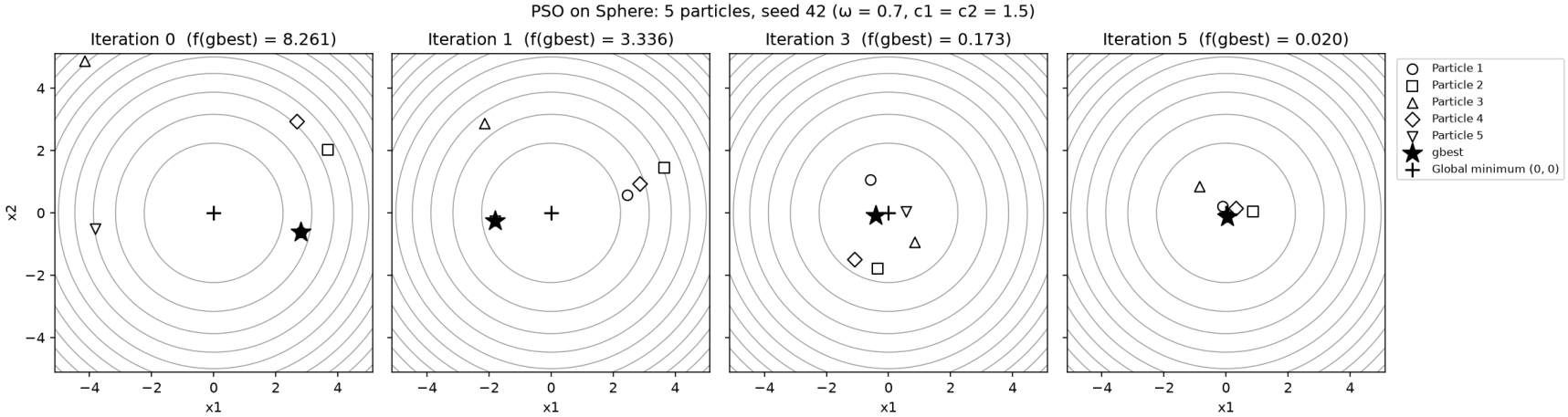



**Cognitive term is 0 at the first step (pbest = x).** One particle gets worse, yet gbest improves from −2 to −1.15.

*Team worked example (report Table 3.2)*

# Five particles, five iterations

Member 2




**f(gbest) by iteration**

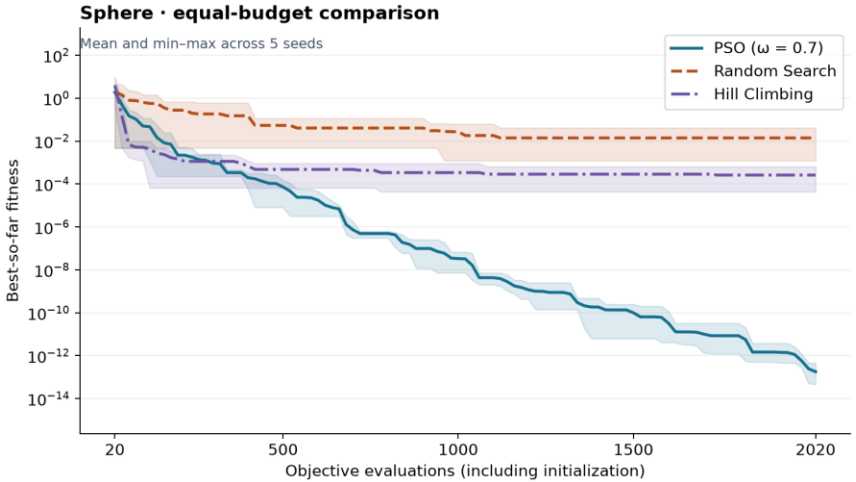
| iter 0 | iter 1 | iter 2 | iter 5 |
|---|---|---|---|
| **8.26** | **3.34** | **0.17** | **0.02** |

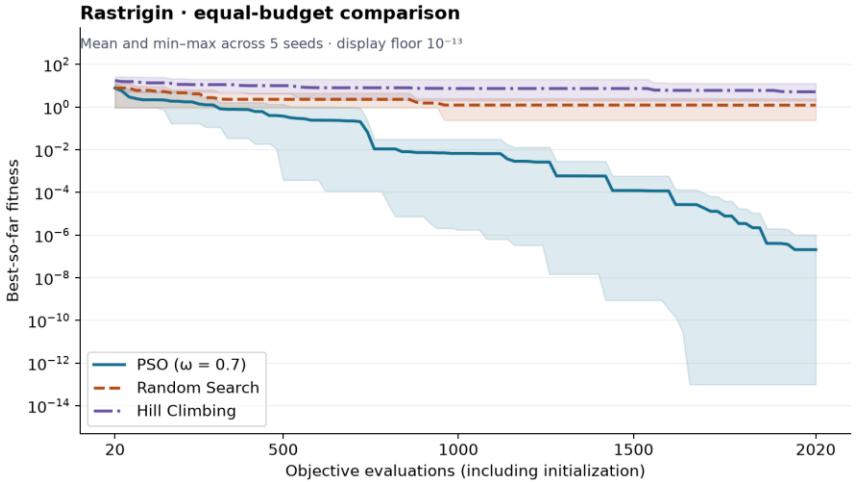
- Iteration 0 → 1: every particle improves
- Velocity clamp (±2) active for 3 particles
- Code reproduces the hand recomputation exactly

*Team experiment: Sphere, 5 particles, seed 42 (report Table 3.3, Figure 3.1)*

# Same budget, two landscapes

Member 3




**42.6 vs 132**
evaluations to reach 0.01 on Sphere: Hill Climbing first

**$1.8 \times 10^{-13}$**
PSO mean final fitness on Sphere, best of the three

**5 / 5 vs 0 / 5**
Rastrigin runs reaching 0.01: PSO vs each baseline

*Team experiments: 2,020 evaluations per run, seeds 0–4 (report Tables 4.1–4.2, Figure 4.1)*

# Inertia depends on the landscape

Member 3

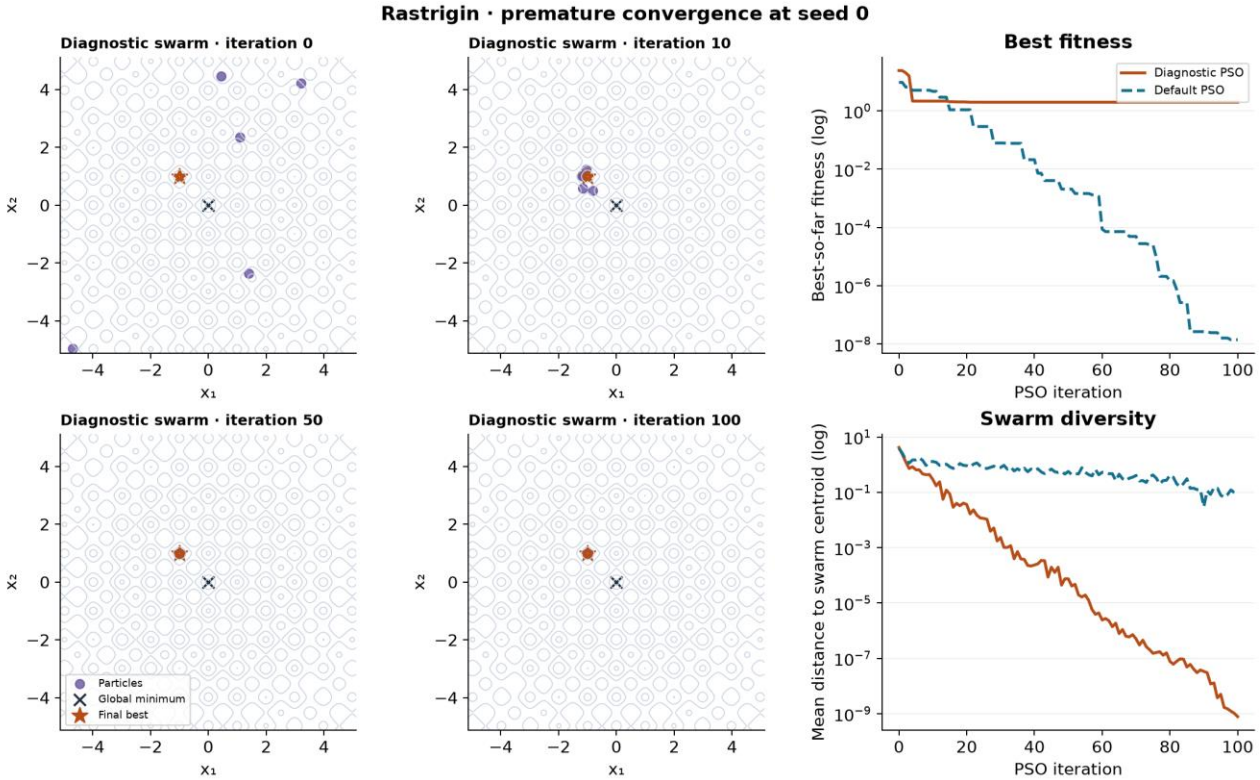
| Mean final fitness | A: ω = 0.4 | B: ω = 0.7 | C: ω = 0.9 |
|---|---|---|---|
| **Sphere** | **$5.5 \times 10^{-31}$** | $1.8 \times 10^{-13}$ | $4.2 \times 10^{-5}$ |
| **Rastrigin** | 0.199 | **$2.1 \times 10^{-7}$** | $8.5 \times 10^{-3}$ |

1. **Sphere:** One basin: strong damping (low ω) settles and refines fastest
2. **Rastrigin:** Four ω = 0.4 runs reach ≈ 0; one stops at 0.995 and dominates the mean
3. **Scope:** Only ω was varied, five seeds each; $c_1$, $c_2$ need their own sweep

*Team experiment: 20 particles, $c_1 = c_2 = 1.5$, seeds 0–4 (report Table 4.3)*

# Where the swarm got stuck

Member 3




## Setting

5 particles, $\omega = 0.4$, $c_1 = 0.5$, $c_2 = 2.5$

## Result

Local minimum at (−0.995, 0.995), $f \approx 1.99$

## Diversity

Mean distance to centroid $4.18 \rightarrow 7.9 \times 10^{-10}$

## Default run

Same seed stays spread out, $f \approx 1.4 \times 10^{-8}$

*Team experiment: Rastrigin, seed 0 (report Figure 4.2)*

# Exploration and exploitation

Member 3

1

**Personal best**
keeps each particle's discoveries and the swarm's diversity

2

**Global best**
focuses the swarm on the best region found so far

3

**Inertia**
decides how far particles carry on past what attracts them

**Too much exploitation** (low inertia, strong social pull, few particles) can collapse the swarm onto a suboptimal basin.

**Applications:** image analysis, power systems and control lead PSO use (Poli et al., 2007), e.g. solar power tracking (Alshareef et al., 2019) and UAV path planning (Phung & Ha, 2021).
**Limits:** two 2-D functions with the optimum at the origin, five seeds, one topology.

# Thank you · Questions?

Kennedy, J. & Eberhart, R. (1995). Particle swarm optimization. Proc. ICNN'95, 1942–1948.

Poli, R., Kennedy, J. & Blackwell, T. (2007). Particle swarm optimization: An overview. Swarm Intelligence 1, 33–57.

Alshareef, M. et al. (2019). PSO for PV power point tracking. Energies 12(4), 623.

Phung & Ha (2021). Spherical vector-based PSO for UAV path planning. Appl. Soft Comput. 107, 107376.

AI tools (Claude, OpenAI Codex) assisted with code and drafting; all results were checked by the team.